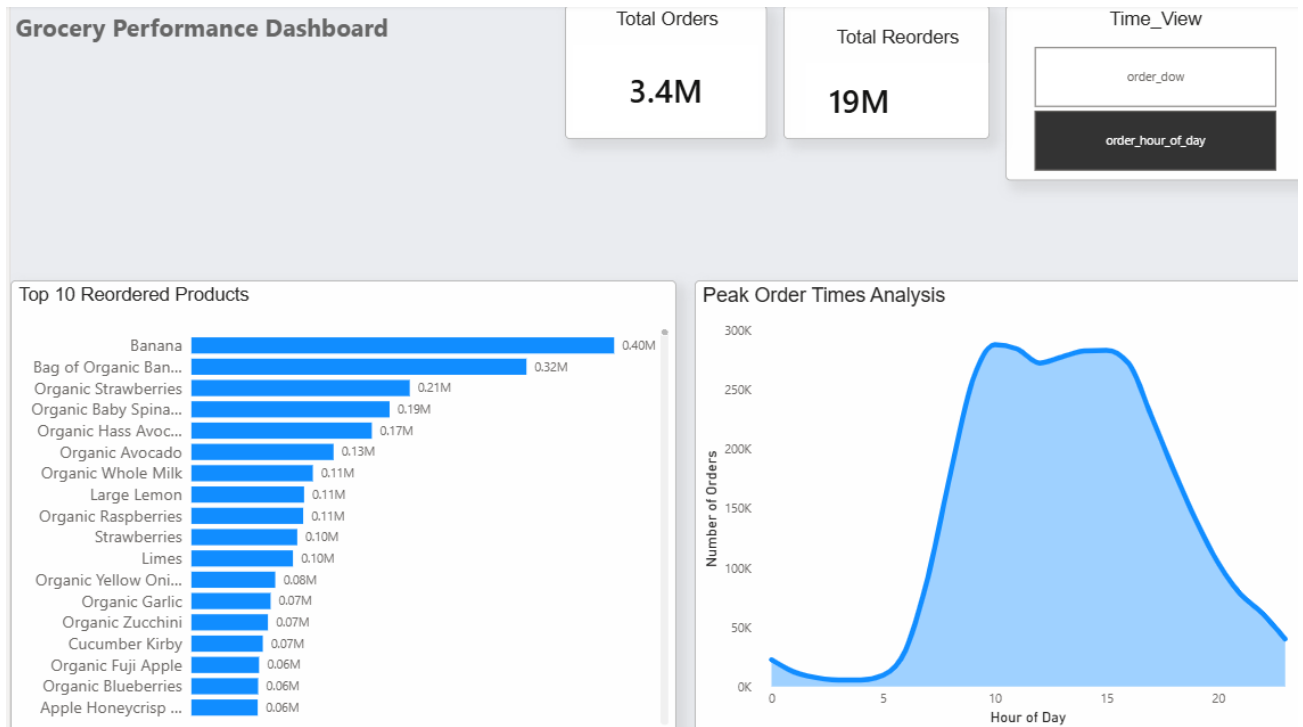


Project: Instacart Grocery Orders & Customer Retention Analysis



Executive Summary & Insights

- **The Challenge:** Analyzed order volume and customer retention using a large grocery dataset. Used Google BigQuery and Power BI to understand customer purchasing behavior and identify peak order times.
- **Key Insight 1 (Peak Hours):** Orders are highest between 10:00 AM and 3:00 PM, with demand remaining high throughout this period.
- **Key Insight 2 (Top Reordered Products):** Fresh products such as Bananas, Organic Strawberries, and Spinach are among the most reordered products.
- **Key Insight 3 (Repeat Purchases):** With over 19M reordered items across 3.4M orders, the data shows strong repeat-purchase behavior among customers.

Business Recommendations:

- **Recommendation 1:** Increase delivery and warehouse capacity during the 10:00 AM - 3:00 PM peak period to reduce delivery delays and handle high demand.
- **Recommendation 2:** Promote fresh and organic products through campaigns, push notifications, and in-app banners to encourage repeat purchases and improve customer retention.
- **Business Impact:** Matching capacity with peak demand and promoting highly reordered products can improve customer retention, increase repeat purchases, and improve operational efficiency.

Technical Execution & Data Pipeline:

- 1. Methodology & Architecture :** To handle millions of rows of transactional data, I used Google BigQuery (SQL) for data analysis and aggregation. I analyzed order patterns, peak order times, and product reorders. The results were then connected to Power BI to create an interactive dashboard and present the key business insights.
- 2. SQL Analysis :** Below are examples of the SQL queries used to analyze peak order times and the most reordered products.

```
-- 1. Order Volume Analysis
-- Calculate the total number of products per order to identify the largest orders.
SELECT
    order_id,
    COUNT(product_id) AS total_order
FROM `data-analysis-projects-496119.instacart_db.order_product`
GROUP BY order_id
ORDER BY total_order DESC;

-- =====

-- 2. Peak Order Times Analysis
-- Determine the distribution of orders throughout the day to identify high-traffic hours.
SELECT
    order_hour_of_day,
    COUNT(order_id) AS total_orders
FROM `data-analysis-projects-496119.instacart_db.orders`
GROUP BY order_hour_of_day
ORDER BY order_hour_of_day ASC;

-- =====

-- 3. Customer Retention & Product Popularity
-- Retrieve the top 10 most frequently reordered products.
SELECT
    p.product_name,
    SUM(op.reordered) AS total_reorders
FROM `data-analysis-projects-496119.instacart_db.order_product` op
JOIN `data-analysis-projects-496119.instacart_db.products` p
    ON op.product_id = p.product_id
GROUP BY p.product_name
ORDER BY total_reorders DESC
LIMIT 10;
```